

Ali Aghababaei

NLP & Generative AI Engineer

Orientation Year Visa | authorized to work in the Netherlands without sponsorship

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Professional Summary

Applied AI Engineer who designs, builds, and ships production LLM systems. Experienced in agentic AI with LangGraph, RAG pipeline engineering, transformer fine-tuning with LoRA/PEFT, and end-to-end deployment across FastAPI, Docker, and cloud infrastructure (Azure, CI/CD, Infrastructure as Code). Proven track record of taking systems from prototype to production with automated evaluation, MLflow experiment tracking, and CI-based quality gates. First co-author on a 2025 arXiv paper on NLP explainability. MSc ICT, University of Padova, 110/110. Based in Rotterdam, authorized to work in the Netherlands without sponsorship.

Technical Skills

Languages	Python, SQL
ML & AI	PyTorch, Hugging Face Transformers, Scikit-learn, TensorFlow, NumPy, Pandas, Deep Learning, Machine Learning, Supervised Learning, Feature Engineering
NLP & GenAI	LangGraph, Agentic AI, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), BERT, RoBERTa, LangChain, Prompt Engineering, Vector Search, sentence-transformers, ChromaDB, LoRA/PEFT, SpaCy, Groq API, Named Entity Recognition (NER), Information Retrieval
Evaluation	RAGAS, LLM Evaluation, A/B Testing, ROUGE, F1, BERTScore, Explainable AI (XAI), Integrated Gradients, Captum, SHAP, Data Curation
Cloud & Infrastructure	Azure (Container Apps, OpenAI, AI Search, ML, Container Registry, Database for PostgreSQL), Terraform, CI/CD (GitHub Actions, OIDC), Docker, FastAPI, Git, pytest, Application Insights, MLOps, MLflow, REST APIs, Cloud Computing, Kubernetes, AWS, Linux, Unix, SQLite

Projects

Property Insights Assistant (LangGraph, PostgreSQL, Azure) [GitHub](#)
Jul 2026

- Built an 8-node LangGraph agent with SQL, RAG, and cancellation-risk routing over PostgreSQL; added a deterministic answer-grounding node, a sampled LLM-as-judge, and structured logging with a human-review queue that promotes flagged cases into a regression suite
- Identified and removed target leakage from a cancellation-risk model, compared 4 model families via cross-validation, and reported an honest **0.706 ROC-AUC**
- Migrated to Azure end-to-end (Container Apps, OpenAI, AI Search, ML) with Terraform and OIDC-secured CI/CD gating **38 pytest tests**; Application Insights tracing shows **p50 29ms** for the risk endpoint

Agentic RAG System (LangGraph, ChromaDB, RAGAS) [Live](#) | [GitHub](#)
Apr 2026

- Built an 8-node agentic AI system with LangGraph: arXiv search with query reformulation loop, session-isolated ChromaDB retrieval with per-paper re-ranking, contradiction detection, and grounded answer synthesis with citations
- Evaluated with RAGAS across 20 queries: **Faithfulness 0.921**, Answer Relevancy 0.885; deployed live on Hugging Face Spaces

Dutch Constitution Q&A: RAG Pipeline with Evaluation [Live](#) | [GitHub](#)
Apr 2026

- Built end-to-end RAG pipeline (PDF ingestion, MiniLM embeddings, ChromaDB, Llama 3.1 via Groq) served

via FastAPI and Gradio; **GitHub Actions CI/CD pipeline** runs 13 pytest tests on every push and auto-deploys to Hugging Face Spaces on success

- Tracked all evaluations in **MLflow**; identified retrieval failures through tracking, replaced character-based chunking with structure-aware Article-level splitting, improving **Faithfulness from 0.55 to 0.86 (+0.31)** via documented A/B comparison

Explainable NLP with Integrated Gradients (BERT, RoBERTa, Captum)

[Paper](#) | [GitHub](#)

Jan 2025

- Deliberately fine-tuned BERT and RoBERTa with controlled overfitting to sharpen class-distinctive token attribution; applied Integrated Gradients via Captum for token-level explainability; benchmarked against DeepLIFT and GradientSHAP; built custom SpaCy rendering pipeline for subword tokenization and negation alignment
- Validated findings with social psychologists; contributed to a published arXiv preprint (2025) as first co-author

Experience

Research Assistant

University of Padova, Italy

Jun 2024 – Dec 2024 · Part-time · Hybrid

- Fine-tuned BERT and RoBERTa on sociopsychological datasets to extract class-distinctive keywords for dictionary expansion and recursive dataset construction; applied Integrated Gradients (Captum) for token-level attribution, benchmarked against DeepLIFT and GradientSHAP
- Built a custom SpaCy rendering pipeline for subword tokenization and negation alignment; findings validated through expert human evaluation by social psychologists
- Contributed to a published arXiv preprint (2025) as first co-author alongside researchers from University of Padova and SWPS University Warsaw

Technical Expert

Mazand Phase Industrial Power Company, Iran

Nov 2019 – Mar 2022 · Full-time

- Designed electrical panels and test methodologies, producing mechanical and electrical drafts in AutoCAD and ePLAN
- Supervised project execution, commissioning, and repair of power systems and electrical inverters
- Delivered product training to clients, translating technical specifications into practical operating procedures

Electrical Engineering Intern

Sorak Taban Electricity Distribution Company, Iran

Jun 2018 – Aug 2018 · Internship

- Assisted in infrastructure planning and technical documentation during undergraduate studies.

Education

M.Sc. in ICT for Internet and Multimedia

University of Padova, Italy

Sep 2022 – Jul 2025

Grade: 110/110 Thesis: *Application of Integrated Gradients Explainability to Sociopsychological Semantic Markers*

Erasmus Exchange Program

Warsaw University of Technology, Poland

Oct 2023 – Feb 2024

Grade: 4.75/5.0

B.Sc. in Electrical Engineering

Ferdowsi University of Mashhad, Iran

Sep 2014 – Sep 2019

Publications

Application of Integrated Gradients Explainability to Sociopsychological Semantic Markers

arXiv preprint (2025) · arxiv.org/abs/2503.04989

Artificial Intelligence in Agro-Food Systems: From Farm to Fork

Foods Journal, MDPI (2025) · doi.org/10.3390/foods14030411

Languages

English (C2, IELTS 8.5) Persian (Native) Italian (A2) Dutch (beginner, learning)